QBioS Laboratory on Outbreaks: From Prediction to Control

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I. OUTBREAKS: FROM DETERMINISTIC MODELS TO STOCHASTIC REALIZATIONS

How do public health scientists use models of outbreaks to forecast the shape of outbreaks to come, to interpret the shape of outbreaks that have already taken place, and/or to determine how best to intervene? This laboratory provides a direct path towards building dynamic simulation approaches to support the control and prevention of infectious disease. In doing so, the laboratory will leverage the core ideas of the textbook to transform epidemic theory into simulation dynamics. This need remains both essential and timeless. These laboratory notes span a five year development process, from the Fall of 2015 to the Fall of 2020. In doing so, they span the Ebola virus disease outbreak in West Africa to the global Covid-19 pandemic. In Fall 2015, it was already apparent that computational models could be used in real-time to support outbreak responses. That is: relatively simple models were used to make projections, guide interventions, and estimate parameters that could connect real-world data and ongoing issues of control. The issues of how to connect population models is ongoing; there have been multiple EVD outbreaks since then, including one in 2018 in the Democratic Republic of Congo, with some concern that these epidemics will become endemic. Nonetheless, there is also optimism, given the development of the 'rVSV-ZEBOV' vaccine that protects against the Zaire strain (the same strain that was the causative agent of the 2014-2015 outbreak). In contrast, the role of theory and simulation in shaping the initial and sustained response to a global Covid-19 pandemic is yet to be written. Models have been critical in shaping responses, but the often simplified choices made have also been criticized, given the need to take actions based on inference from intentionally simplified or incomplete representations of the real world.

Indeed, it is critical to keep in mind that variants of the kinds of models developed in this laboratory are used, in practice, by academic modeling groups, governmental agencies (like the WHO and CDC), and independent non-governmental organizations to help policy makers decide how best to allocate resources, design vaccination programs, public health campaigns, and other interventions to prevent and control disease. Indeed, one could argue that despite their simplicity, this laboratory actually goes above and beyond some intervention-level tools. For example, the initial 'Ebola Response Modeling tool' developed by the CDC to provide real-time feedback and scenario response was not a program written in Python, MATLAB, C, but instead a Microsoft Excel spreadsheet. The tool was announced via the CDC `https://www.cdc.gov/media/releases/2014/s0923-ebola-model-Factsheet.html` and is available for inspection here: `https://stacks.cdc.gov/view/cdc/24900`. You might wonder: why an Excel spreadsheet? Answering such a question would take time and steer us away from the pressing issues in this laboratory, but suffice it to say that the kinds of models developed here have greater flexibility but do require that one knows how to program in order to manipulate them. From a scientific perspective, there is a deeper advantage, the core equations here include feedback between different types of individuals, whether susceptible, infectious, recovered, or removed, as well as potential interventions to move individuals from one state to another (e.g., via safe burial) or to change interaction rates between individuals (e.g., via quarantine). By using a high-level programming language and not a spreadsheet, we are able to develop a suite of flexible approaches including dynamic and stochastic models. These models are also more robust – technically. The state of modeling in response to the Covid-19 pandemic reflects a maturation of real-time response efforts: including epidemic models, model-data integration, scenario building, real-time estimators, and more. This lab can't do all of that. But, it will provide the core principles to help you move beyond armchair epidemiology status. The core methods here form the basis for state of the art response models found in the rapid proliferation and interest in public health globally.

In brief, this computational laboratory will teach you how to (i) model an epidemic outbreak; (ii) explore the relationship between speed, strength, and size of an outbreak; (iii) develop a stochastic simulation of disease dynamics, flexible enough to be applied to real-world cases. This is a lot - but it builds on concepts you've already explored earlier in the class. As always, you can do it!

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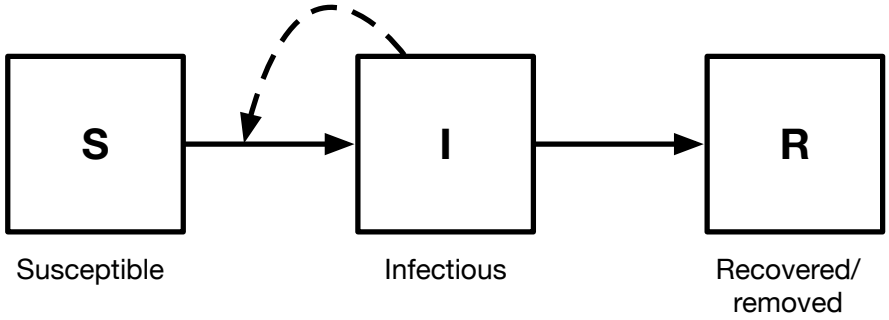


FIG. 1: Disease dynamics represented in terms of susceptible, infectious, and recovered/removed individuals, i.e., a SIR model. Infectious individuals can come into contact with susceptible individuals leading to new infections. Infectious individuals recover or are removed from circulation (in the case of potentially fatal diseases). These interactions form the basis for the rates and dynamics in both deterministic and stochastic versions of population-level epidemic models.

II. EPIDEMIC MODEL - FUNDAMENTALS

A. Transitions to Outbreaks

Consider an infectious diseases that can spread in a population of size N. Individuals may be susceptible, infectious, or recovered/removed. We term these individuals S, I and R respectively, such that $S + I + R = N$. The re-scaled dynamics can be written as

$$\begin{aligned} \dot{S} &= -\beta SI \\ \dot{I} &= \beta SI - \gamma I \\ \dot{R} &= \gamma I \end{aligned}$$

given the two processes, infection at a rate βSI and recovery at a rate γI. The main text includes the full derivation of these equations and the meaning of β. Note also that variability in susceptibility can lead to fundamental changes in these equations. Here, this model denotes the fraction of individuals of each type, i.e., $S + I + R = 1$. Simulations of this model should use the following core code, beginning with the representation of the ODE:

```
function dydt = sir_model(t,y,pars)
% function dydt = sir_model(t,y,pars)
% SIR Model

S=y(1);
I=y(2);

% The model
dSdt = -pars.beta*S*I;
dIdt = pars.beta*S*I-pars.gamma*I;
dRdt = pars.gamma*I;

dydt = [dSdt; dIdt; dRdt];
```

Applying this model requires a relevant choice of epidemiological parameters. Consider susceptible individuals that interact with c individuals per unit time, of which I/N are infectious, and of which a fraction p of such contacts lead to a new infectious event. In that case we expect the rate of transmission per susceptible individual to be cpI/N multiplied by the number of susceptible individuals to yield $cpSI/N$ or $\beta \equiv cp$.

Challenge Problem: Outbreak Criteria

Complete the simulation code below for the spread of an infectious disease beginning with 1 individual out of 10000 and estimate the value of $\mathcal{R}_0$. Do you expect the disease to spread or not? Then, change the value of p to 0.01 – will the disease spread, why or why not?

```
% main data goes here
pars.c = 20;      % Contacts per unit time (e.g., days)
pars.p = 0.025;   % Probability of infectious contact
pars.beta = ...   % Transmission rate
pars.gamma = 1/4; % Recovery rate (days^-1)
pars.basR0 = ...  % Basic reproduction number
pars.N = 10000;
pars.I0= 1;
pars.S0= pars.N-pars.I0;

% Run the model
[t,y]=ode45(@sir_model,[0 100],[pars.S0 pars.I0 0]/pars.N,[],pars);

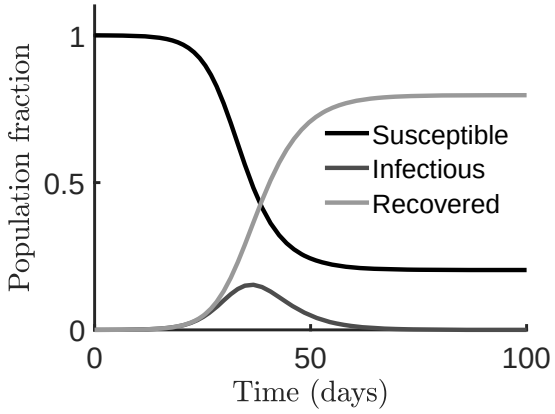
% Plot the results
tmph=plot(t,y);
set(tmph,'linewidth',3);
xlabel('Time (days)');
ylabel('Population fraction');
tmplh = legend('Susceptible','Infectious','Recovered');
legend('boxoff');
```

Solutions to Challenge Problem: Outbreak Criteria

The missing components are shown below.

```
pars.beta = pars.c*pars.p;  % Transmission rate
pars.basR0 = pars.beta/pars.gamma; % Transmission rate * transmission period
```

These parameters describe an outbreak for the SIR model given $\mathcal{R}_0 = 2$, $\beta = 0.5$ days^{-1}, $\gamma = 0.25$ days^{-1}. As a result, the simulation of the outbreak looks as follows:



In this case, the outbreak occurs because $\mathcal{R}_0 > 1$. However, when $p = 0.01$ then $\beta = 0.2$ and $\mathcal{R}_0 = 0.8$ which is less than 1. A simulation (results not shown) leads to the recovery of infectious individuals without a net increase in the number of cases.

B. Speed, Strength, and Size

Once an outbreak starts, how far will it go? That is to say: how many individuals in a population will become sick? The relationship between speed, strength, and size is critical to understand outbreak dynamics. Let's define each term. The speed can be measured in terms of the rate, r, of exponential growth of the outbreak, i.e., $I(t) \sim e^{rt}$. The strength can be measured in terms of the dimensionless number $\mathcal{R}_0$. This value is known as the basic reproduction number. It denotes the average number of new infections caused by a single infectious individual in an otherwise susceptible population. Finally, the size of the outbreak can be measured in terms of the fraction of the population infected at the culmination of the outbreak, $R(t \to \infty)$, which is equivalent to $1 - S(t \to \infty)$. The main text presents a derivation of the strength-size relationship, specifically:

$$\mathcal{R}_0(S_\infty - 1) = \log(S_\infty) \tag{1}$$

as well as the strength-speed relationship, i.e.,

$$r = \gamma(\mathcal{R}_0 - 1) \tag{2}$$

Here, you will explore both of these systematically, gaining an intuition for these two relationships.

1. Strength and Speed

It would seem that outbreaks with higher values of $\mathcal{R}_0$ would lead to a faster increase in case counts. But that is not necessarily the case. One way to explore this is to modulate both β and γ, i.e., the transmission and recovery rates, which influence both strength and speed. To begin, let's first measure the speed in an outbreak. The code below simulates an outbreak using $\mathcal{R}_0 = 2$, $\beta = 0.5$ days^{-1}, and $\gamma = 0.25$ days^{-1}. For this combination, we expect $r = 0.25$ days^{-1}. The following code illustrates the following steps. First, simulate the model. Second, fit a line to the log-transformed infectious counts, $I(t)$. Note that you could also find the line of best fit to the cumulative case incidence, $I(t) + R(t)$. The slope of this line should be the estimated $\hat{r}$.

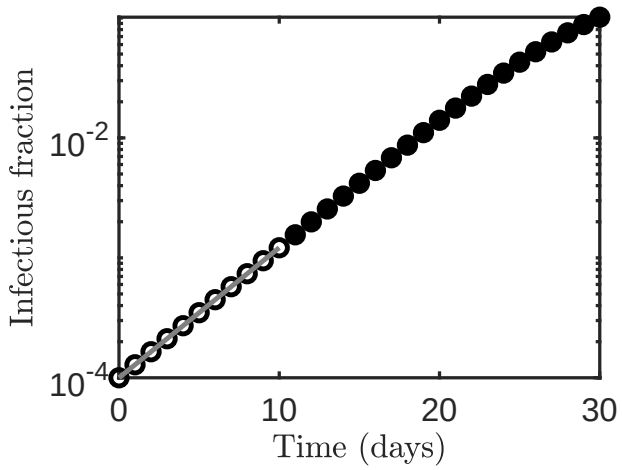
```
% Run the model over 10 days
[t,y]=ode45(@sir_model,[0:1:10],[pars.S0 pars.I0 0]/pars.N,[],pars);

% Find the slope
[p,s]=polyfit(t,log(y(:,2)),1);

% Plot the data and overlay the best-fit exponential
tmph=semilogy(t,y(:,2),'ko');
set(tmph,'linewidth',3);
hold on
tmph=semilogy(t,exp(p(1)*t+p(2)),'r-');
set(tmph,'linewidth',3);

% Use solid points for the future k
[t,y]=ode45(@sir_model,[0:1:30],[pars.S0 pars.I0 0]/pars.N,[],pars);
tmpi=find(t>10);
tmph=semilogy(t(tmpi),y(tmpi,2),'ko');
set(tmph,'linewidth',3,'markerfacecolor','k');
```

When implemented, this code results in the following image:



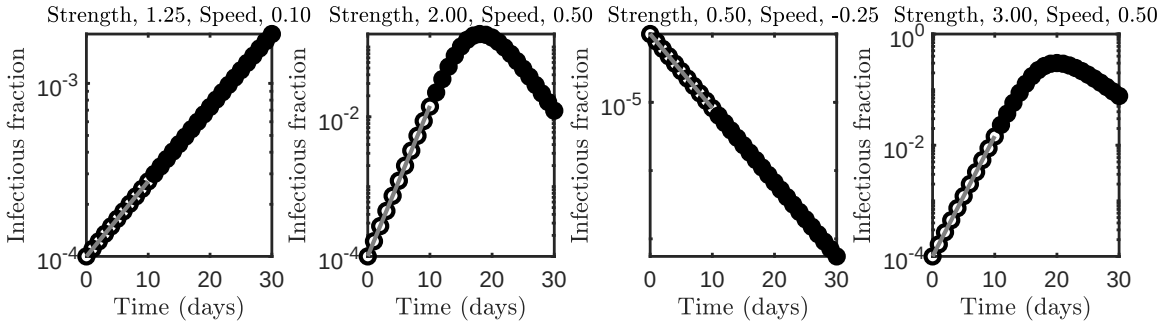
As is apparent the fraction of infectious individuals does grow exponentially to begin, at precisely the rate expected in theory. In this case, the slope of the line is 0.2496 days^{-1} and the expected value is $r = 0.25$ days^{-1}. With this code snippet in mind, your next challenge is to examine the speed-size relationship given variation in β and γ.

Challenge Problem: Strength-Speed Relationships

Determine the strength and speed for the following sets of disease parameters

Transmission β	Recovery γ	Strength $\mathcal{R}_0$	Speed r
0.5	0.4	...	...
1	0.5	...	...
0.25	0.5	...	...
0.75	0.25	...	...

Solutions to Challenge Problem: Strength-Speed Relationships.
The solution proceeds as before, by simulating the dynamics, fitting an exponential and then comparing to the theoretically expected value. Notably the third case has $\mathcal{R}_0 < 1$ which means the speed is negative, i.e., the number of cases declines over time. Altogether, the images look as follows:



for which the estimated values of strength and speed are:

Transmission β	Recovery γ	Strength $\mathcal{R}_0$	Speed r
0.5	0.4	1.25	0.1
1	0.5	2	0.5
0.25	0.5	0.5	-0.25
0.75	0.25	3	0.5

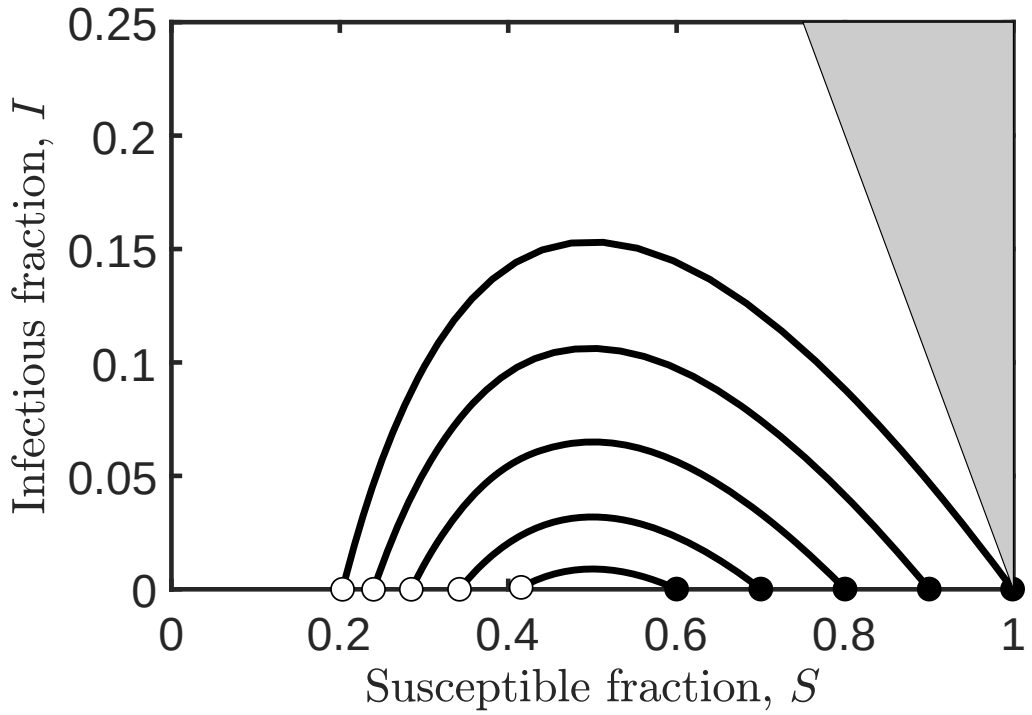
2. Strength and Size

An outbreak is not expected to infect all individuals. The final size of the epidemic can be measured in terms of the cumulative case count, i.e., the value of $R_\infty = 1 - S_\infty$. The next code set shows how to systematically modify the initial fraction of infected. In these overlaid simulations, the initial values are marked by black circles (on the right side of each arc) and the final points by gray circles (on the left of each arc). The dynamics in the phase suggest there is a critical value of $1/\mathcal{R}_0$ where an initial outbreak will not spread. This observation is the basis for findings that not all individuals in population must be vaccinated for a vaccine to be effective at the population level! However, an ancillary finding is that the fraction of the population that should be vaccinated increases with the strength of the disease, i.e., measured in terms of $\mathcal{R}_0$.

```
% Modify the initial values
pars.N = 10000;
pars.S0_range=[0.6 0.7 0.8 0.9 0.999];
pars.I0_range=1/pars.N*ones(1,5);
pars.R0_range=1-pars.S0_range-pars.I0_range;

% Run the model
for i=1:length(pars.S0_range),
  [t,y]=ode45(@sir_model,[0:1:200],[pars.S0_range(i) pars.I0_range(i) pars.R0_range(i)],[],pars);
  tmph=plot(y(:,1),y(:,2),'k-');
  set(tmph,'linewidth',3);
  hold on
  tmph=plot(y(end,1),y(end,2),'ro');
  set(tmph,'markerfacecolor','r');
  tmph=plot(y(1,1),y(1,2),'ko');
  set(tmph,'markerfacecolor','k');
end
% Show the excluded regime
patch([1 1 0.75 1],[0 0.25 0.25 0],[0.8 0.8 0.8]);
```

Running this code yields the following trajectories where the shaded region denotes inaccessible parts of phase space given that $S + I \leq 1$.



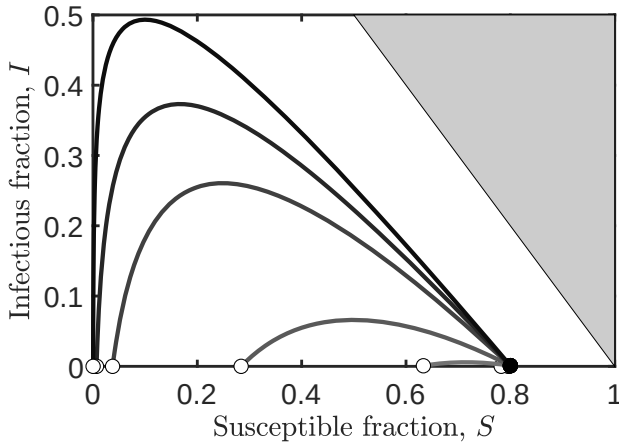
This example of visualizing outbreak dynamics in the phase plane also provides the basis for the next challenge problem.

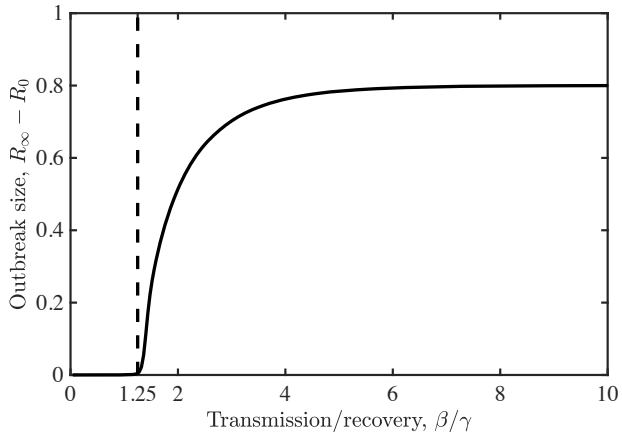
Challenge problem: Strength-Size Relationships in Phase Space

First, fix the value of the recovery rate to $\gamma = 0.25$, then modulate the transmission rate from $\beta = 0.25$ to $\beta = 2.5$. Assume that initially, 80% of the population is susceptible and 20% is recovered. In this case, does the disease always spread? Why or why not? In addition, find the outbreak size and show how it relates to $\mathcal{R}_{eff}$ – the effective reproduction number given the initial susceptible population.

Solutions to Challenge problem: Strength-Size Relationships in Phase Space

The effective reproduction number $\mathcal{R}_{eff} = \frac{\beta S_0}{\gamma}$ which ranges from 0.8 to 8 in this case. Hence, even though $\beta > \gamma$, there will not be an outbreak until $\beta > \gamma/S_0$ or $\beta_c = 5/16 = 0.3125$. This is apparent in the plots below (left-phase space; right-relationship between final size and β). As noted in the right plot, the ratio of β/γ must exceed 1/0.8 or 1.25 for the outbreak to initiate. Note that analytical derivations of the strength-size relationship given a partially susceptible population is presented in the main text.





Solutions to Challenge problem: Strength-Size Relationships in Phase Space – Continued

The code to generate the phase plots is as follows, noting that the labels are not included and should be added to enhance interpretation and readability:

SI phase dynamics

```
pars.N=10000;
pars.beta_range=[0.25 0.3 0.35 0.5 1 1.5 2.5];
pars.gamma = 1/4;
pars.S0=0.8;
pars.I0=1/pars.N;
pars.R0=1-pars.S0-pars.I0;

% Run the model
for i=1:length(pars.beta_range),
  pars.beta=pars.beta_range(i);
  pars.effR0(i) = pars.beta/pars.gamma*pars.S0;
  pars.r(i)=pars.gamma*(pars.effR0(i)-1);
  [t,y]=ode45(@sir_model,[0:0.1:200],[pars.S0 pars.I0 pars.R0],[],pars);
  tmph=plot(y(:,1),y(:,2),'k-');
  set(tmph,'linewidth',3,'color',[0.75 0.75 0.75] - [0.1*i 0.1*i 0.1*i]);
  hold on
  tmph=plot(y(end,1),y(end,2),'ko');
  set(tmph,'markerfacecolor','k');
  tmph=plot(y(1,1),y(1,2),'ko');
  set(tmph,'markerfacecolor','w');
end
% Show the excluded regime
patch([1 1 0.5 1],[0 0.5 0.5 0],[0.8 0.8 0.8]);
```

The relationship between transmission and size

```
% main data goes here
pars.N=10000;
pars.beta_range=[0.01:0.01:2.5];
pars.gamma = 1/4;
pars.S0=0.8;
pars.I0=1/pars.N;
pars.R0=1-pars.S0-pars.I0;

% Run the model
for i=1:length(pars.beta_range),
  pars.beta=pars.beta_range(i);
  pars.effR0(i) = pars.beta/pars.gamma*pars.S0;
  pars.r(i)=pars.gamma*(pars.effR0(i)-1);
  [t,y]=ode45(@sir_model,[0:0.1:200],[pars.S0 pars.I0 pars.R0],[],pars);
  pars.size(i)=y(end,3)-pars.R0;
end
tmph=plot(pars.beta_range/pars.gamma,pars.size,'k-');
set(tmph,'linewidth',3);
hold on
tmph=plot([1/pars.S0 1/pars.S0],[0 1],'k--');
set(tmph,'linewidth',3);
set(gca,'xtick',[0 1/pars.S0 2 4 6 8 10]);
```

C. From Outbreaks to Endemics

The SIR model, with rapid loss of immunity can be written in terms of an SI model:

$$\dot{S} = -\beta SI + \gamma I$$
$$\dot{I} = \beta SI - \gamma I$$

This model assumes that there is no immunity. Instead, recovered individuals are immediately susceptible. As a restul, the dynamics include a new long-term possibility: an endemic disease state where $S^* = \frac{1}{\mathcal{R}_0}$ and $I^* = \frac{\mathcal{R}_0 - 1}{\mathcal{R}_0}$. This SI model can be coded as follows:

```
function dydt = si_model(t,y,pars)
% function dydt = si_model(t,y,pars)
% SI Model

S=y(1);
I=y(2);

% The model
dSdt = -pars.beta*S*I+pars.gamma*I;
dIdt = pars.beta*S*I-pars.gamma*I;

dydt = [dSdt; dIdt];
```

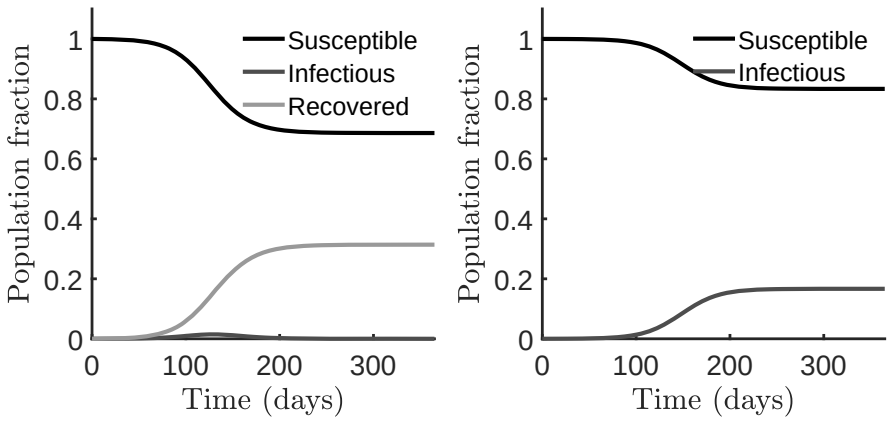
As should be apparent, there is only an endemic state with $I^* > 0$ when $\mathcal{R}_0 > 1$.

Challenge problem: From Epidemic Outbreaks to Endemics

Using $\beta = 0.3$ and $\gamma = 0.25$ days^{-1} show the convergence of the dynamics to an endemic state, i.e., from an outbreak to persistent infectious cases. And then compare and contrast this with dynamics in which immunity is permanent. For these parameters, simulate the dynamics over a 1 year = 365 day period.

Solutions to Challenge problem: From Epidemic Outbreaks to Endemics

The contrast between the SIR model and the SI model can be seen in the plots below of the contrasting outcomes. In both cases there is an outbreak. However, with $\mathcal{R}_0 = 1.2$, we expect 83.3% of the population to be susceptible at equilibrium, and therefore 16.7% to be infected – precisely as observed.



III. STOCHASTIC EPIDEMICS

A. Definition of the Model

Stochastic realizations of the SIR model are simulated using the Gillespie framework, given the "reaction" events in the following table:

Process	**Reaction**	**rate/probability**
Infection	$S + I \to 2I$	$r_1 = \beta_I S \frac{I}{N}$
End of infectiousness (recovery)	$I \to R$	$r_2 = \gamma I = I/T_I$

These processes denote transitions amongst individuals who are susceptible (S), infectious (I), and recovered/removed (R). The total population is fixed at $N = S + I + R$. Epidemics are initiated with one infectious individual in an otherwise susceptible population. Mathematically, the initial state is $\mathbf{y} = (N_0 - 1, 1, 0)$ at $t = 0$, where the ordering of terms is susceptible, infectious and recovered. The total rate of outbreak-associated events is $r_{tot} = \sum_{i=1}^{2} r_i$. The time until next event is determined randomly such that $\delta t \sim \frac{-\log \chi}{r_{tot}}$ where χ is a uniformly distributed number between 0 and 1. In this way, the time between events follows an exponential distribution with rate r_{tot}. Then, the probability of each event is r_i/r_{tot}. The process stops when $I(t) = 0$, as in that case $r_{tot} = 0$. After selecting an event and updating the discrete number of individuals, the reaction rates are recalculated and the process continues. The same framework can be extended to include other classes (e.g., exposed individuals) as well as other kinds of processes (e.g., post-death transmission in the case of EVD). Trajectories are complete when the epidemic dies out because there are no more infectious individuals. In the present context, we are interested in those trajectories that do not die out before the end of the simulation time.

B. Simulating a stochastic outbreak

In earlier labs, we developed Gillespie models of gene expression. The same principles can be applied here. Here, your objective is two-fold, first to modify the code below to simulate a stochastic epidemic, and then to simulate and compare results from stochastic simulations to your deterministic trajectories.

Challenge Problem: Stochastic SIR Model

Modify the following code to implement a stochastic SIR model including both transmission and recovery events in a population of size N.

```
function [t,y] = stochsim_SIR(trange,y0,pars)
% function [t,y] = stochsim_SIR(trange,y0,pars)
%
% Simulates an SIR model via the Gillespie algorithm
% from t0 to tf in trange given initial
% conditions in y0 = [S0 I0 R0] and parameters
% in pars.  Returns time and values

% Conditions
t0=trange(1);
tf=trange(2);
t(1)=t0;
y(1,:)=y0;
tcur=t0;
ycur=y0;
ind=1;

% Model
while (tcur<tf)
  % Check to see if there is an infection
  if (ycur(2)==0)
    ind=ind+1;
    t(ind)=tf;
    y(ind,:)=ycur;
    break;
  end
  % Rates
  infrate = ...
  recrate = ...
  totrate = infrate + recrate;
  dt = -1/totrate*log(rand);
  tcur=tcur+dt;
  % Event type
  if (rand<(infrate/totrate)) % infection
    ...
    ...
  else % recovery
    ...
    ...
  end
  ind=ind+1;
  t(ind)=tcur;
  y(ind,:)=ycur;
end
```

Solutions to Challenge Problem: Stochastic SIR Model
There are two key components to modify. First, the underlying transition rates must be modified. Those include the infection and recovery rates. For a generalized model there may be other rates as well. Here the correct code is:

```
% Rates
  infrate = pars.beta*ycur(1)*ycur(2)/pars.N;
  recrate = pars.gamma*ycur(2);
% Event type
  if (rand<(infrate/totrate)) % infection
    ycur(1)=ycur(1)-1;
    ycur(2)=ycur(2)+1;
  else % recovery
    ycur(2)=ycur(2)-1;
    ycur(3)=ycur(3)+1;
  end
```

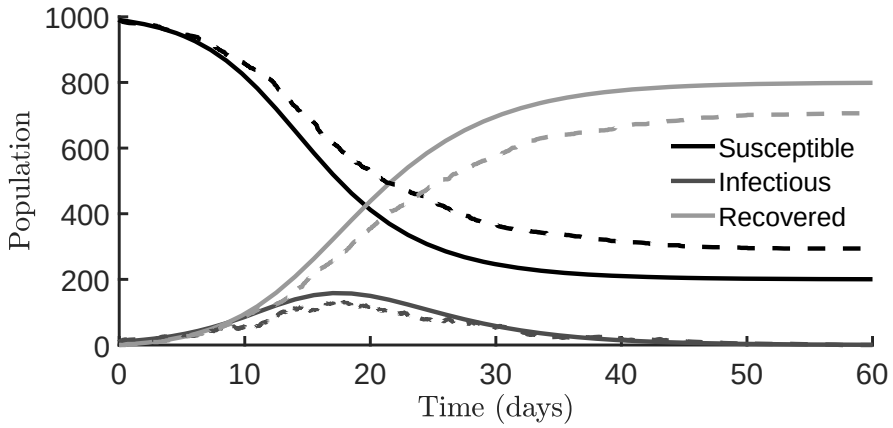
The key here is that there are multiple state variables to change, and they go up/down by 1 depending on whether there is a new infection or a recovery event.

Once you have a working code, simulate it using the following parameters including an initial seed of 10 infectious

individuals. This is usually enough to ensure the disease spreads. The following parameters should provide both enough context and information to intialize the simulation.

```
pars.c = 20;    % Contacts per unit time (e.g., days)
pars.p = 0.025;         % Probability of infectious contact
pars.beta = pars.c*pars.p;  % Transmission rate
pars.gamma = 1/4; %
pars.tf=60;
pars.basR0 = pars.beta/pars.gamma;
pars.N = 1000;
pars.I0= 10;
pars.S0= pars.N-pars.I0;
```

and compare directly to the SIR model which should yield the following kind of result. Here, both the stochastic and deterministic trajectories are plotted against each other.



As is apparent, there can be many differences between a deterministic model and a stochastic trajectory. Understanding that difference is at the very heart of inferring and predicting disease trajectories in real systems. If you have time, you could even try to estimate the speed of the outbreak for a set of trajectories... you will find that there is variability even without considering observational noise or process noise, and that such variation can be substantial at the outset of an outbreak when estimates are most needed!

Finally, it may seem difficult to compare the models, but it is easier than it seems. The key is to run both models but then recall that the ODE model dynamics are in terms of fractions, which must be re-scaled. Here is how to do it:

```
% Run the ODE model
[t,y]=ode45(@sir_model,[0 pars.tf],[pars.S0 pars.I0 0]/pars.N,[],pars);
tmph=plot(t,pars.N*y);
set(tmph,'linewidth',3);
hold on

% Run the stochastic model
[tsim,ysim]=stochsim_SIR([0 pars.tf],[pars.S0 pars.I0 0],pars);
tmphsim=plot(tsim,ysim,'--');
set(tmphsim,'linewidth',3);

% Color the lines the same way (where dashes are for stochastic)
for i=1:3,
  set(tmphsim(i),'color',get(tmph(i),'color'));
end
```